COURSE OVERVIEW

GEOSPATIAL REPRESENTATION LEARNING

Introduction and Kick-off

Winter Semester 2026/27
Mobile Robotics, Geodesy and
Geoinformation & Geodetic Engineering

PRESS → OR SPACE. →

© Marc Rußwurm
Licensed under **Creative Commons Attribution-NonCommercial 4.0 International** (CC BY-NC 4.0).
You may share and adapt these slides for teaching, research, and other non-commercial educational purposes with attribution.
Commercial training, paid workshops, consulting seminars, or incorporation into commercial course products require prior permission.

UNIVERSITÄT BONN

Three Phases:

Phase 1: Lectures & Labs

Build conceptual and technical foundations through lectures and hands-on exercises.

Examination: final written exam on the course content.

Phase 2: Paper Analysis

Analyze and review recent research papers in student groups.

Examination: pass/fail paper presentation with questions.

Phase 3: Group Projects </>

Develop code and conduct a research-oriented geospatial analysis project in groups.

Examination: four-page scientific paper.

Bloom's cognitive levels across the course

Remember	Understand	Apply	Analyze	Evaluate	Create

Timeline at a Glance

Fridays: Lectures 10:00–12:00 in HS VIII (Meckenheimer Allee 172)*
Labs 13:00–15:00 in Seminarraum 2.003 (Nußallee 1)

* Lectures 2 (October 30) and 7 (December 11) are in HS IX (same floor).

Course week	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Event date	Oct 16	Oct 23	Oct 30	Nov 6	Nov 13	Nov 20	Nov 27	Dec 4	Dec 11	Dec 18	Christmas + Jan 8	Jan 15	Jan 22	Jan 29	Feb 5
Phase 1: Lectures	Intro	L1	L2*	L3	L4	No class	L5	L6	L7*	Guest					
Phase 1: Labs		Lab 1	Lab 2	Lab 3	Lab 4										
Phase 2: Papers							P1	P2	P3	P4					
Phase 3: Projects												Group Projects			

Presence and Examination Events

Paper Presentations pass/fail
Nov 27, Dec 4, 11 & 18, 2026

Phase 1 Written exam
February 5th, 2027

Phase 3 paper deadline
Date: TBD

Phase 1: Lectures and Labs

Lectures

10:00–12:00 (s.t.)

HS VIII (Meckenheimer Allee 172)
Lectures 2 (October 30) and 7 (December 11): HS IX (same floor).

Presentation slides introduce the core concepts, terminology, and research context.

They provide the shared conceptual foundation for the hands-on work.

Labs

13:00–15:00

Seminarraum 2.003 (Nußallee 1)

Interactive activities reinforce the lecture through self-directed, collaborative, and hands-on learning:

- Explore representations using PCA and t-SNE
- Collect, clean, and combine geodata
- Train and evaluate machine-learning models
- Investigate model architectures and explain them through peer teaching

Phase 1: Overview - Lectures & Labs

Lecture 1

Representations
What is a representation?

Lecture 2

Geospatial Data
What is geodata?

Lecture 3

Data Representations
Describing places as feature vectors

Lecture 4

Machine Learning on Data Representations
Learning on fixed representations

Lecture 5

Deep Representation Learning I
Learning the representation itself

Lecture 6

Deep Representation Learning II
Learning beyond memorization

Lecture 7

Earth Embeddings
Giving location a meaning

Lab 1

Visualizing high-dimensional representations

Lab 2

Geodata Scavanger Hunt

Lab 3

Machine Learning on Geospatial Data.

Lab 4

Practical: Supervised Deep Learning

Guest Lecture: Frank Feng on TESSERA

December 18, 10:00–12:00 in HS VIII (lecture slot)

Zhengpeng (Frank) Feng
PhD student, Energy and Environment Group
Computer Science, University of Cambridge

TESSERA: *Temporal Embeddings of Surface Spectra for Earth Representation and Analysis* (CVPR 2026)

- Self-supervised foundation model that encodes Sentinel-1/2 **time series** per pixel
- Open, precomputed **global embeddings at 10 m**, annually 2017–2025
- Request a bounding box via **GeoTessera**, then train lightweight models on the embeddings
- TESSERA v2: scaling study and distilled Matryoshka embeddings

geotessera.org · arXiv:2506.20380 · frankfeng-23.github.io

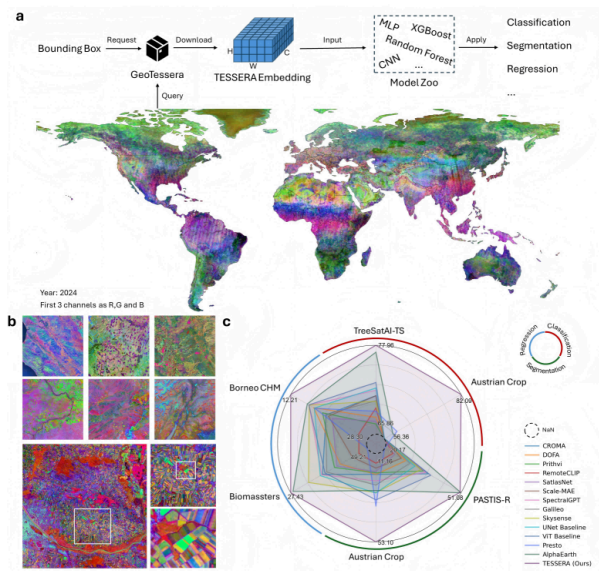


Figure: Feng et al., TESSERA (CVPR 2026)

Phase 2: Paper Analysis

Paper Analysis slides
Open the session deck

PowerPoint presentation template
Coming soon

Format

Each topical session introduces a new research area and assigns papers to the student groups.

1. **Read (between classes):** Study the assigned paper with your group.
2. **Analyze (between classes):** Identify the problem, method, evidence, limitations, and impact, and prepare your presentation.
3. **Present (in class):** Present your analysis, answer questions, and discuss the paper with the class.

Four Topics

Session	Topic
1	Implicit Neural Representations and Approximation
2	Location Encoders
3	Self-Supervised Representation Learning
4	Geospatial Embeddings

Phase 3: Group Projects

Groups of five · One research question · One four-page scientific paper

Solar Fields

Map solar expansion on farmland.

Border Landscapes

Compare farming landscapes across European borders.

Plastic Currents

Explore potential plastic sources and sinks along the Rhine.

Earth Economies

Connect landscape embeddings with production and trade.

Embedding Seasons

Detect double versus triple cropping in southern Brazil.

Generative Fields

Generate satellite imagery from field maps in India.

Paper: Question & hypothesis → Experiment & baselines → Results & limitations
Reproducibility: GitHub repository with code and a short README.

Project descriptions and evaluation rubric ↗

AI policy

The goal of this course is for you to think.

You are encouraged to use ChatGPT, Claude, Gemini, and other AI tools to:

- Retrieve information and explore new ideas
- Structure **your own thoughts**
- Discuss concepts with an LLM agent to sharpen your understanding

Labs and exercises are designed to be interactive, and the exam is handwritten. AI agents cannot do the thinking or complete these activities for you.

Who's in the Room?

Teachers

Marc Rußwurm

Professor

Sawssen Hannachi

Doctoral Researcher

Gabriele Tijunaityte

Doctoral Researcher

What about you? - Live Understanding Pulse

Throughout this course, I will show this widget to touch base on your level of understanding. It synchronizes across all devices, including in-class and virtual participants.

Try it out - click on the line:

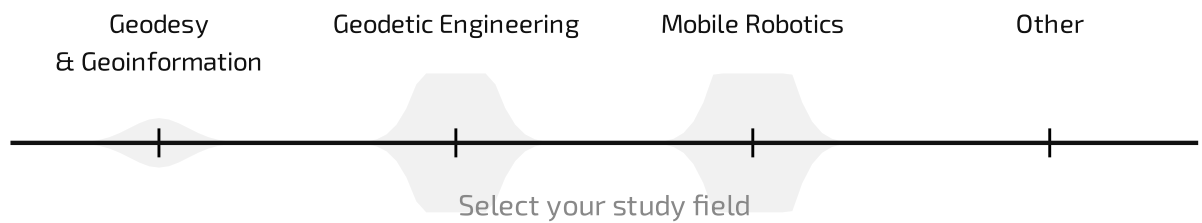
I am completely lost — I fully understand

Click on the line to vote.

This gives me feedback during class and also tracks the general understandability of the slides, so I can improve them in future iterations.

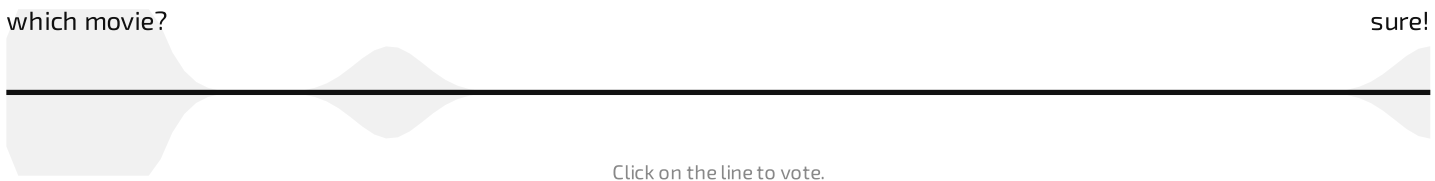
What is your study field?

Click the category that best describes your study programme.



Light gray fill: responses of the 14 participants on 16 October 2026 (10:00–12:00), last vote per person.

I can train Transformer models

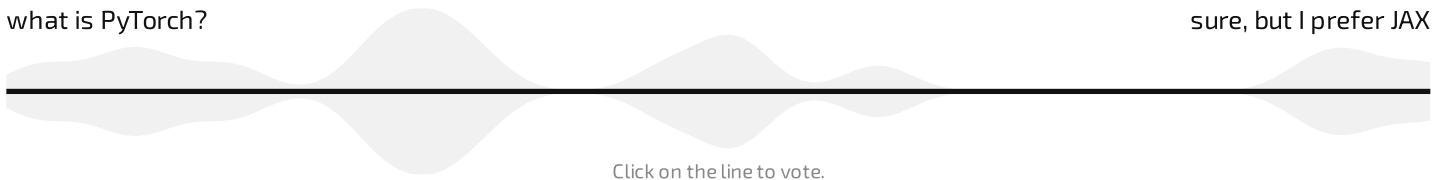


Light gray fill: responses of the 9 participants on 16 October 2026 (10:00–12:00), last vote per person.

I can write a PyTorch data loader

what is PyTorch?

sure, but I prefer JAX



Click on the line to vote.

Light gray fill: responses of the 16 participants on 16 October 2026 (10:00–12:00), last vote per person.

I can explain coordinate reference systems

coordinate?

sure, which one? WGS84?

Click on the line to vote.

Light gray fill: responses of the 15 participants on 16 October 2026 (10:00–12:00), last vote per person.

Course Learning Outcomes

Explain geospatial representations and embeddings, including their potential and limitations.

Compare and curate geospatial datasets from imagery, time series, GIS, and socio-economic sources.

Implement Python ML workflows with fixed and learned geospatial representations.

Evaluate models for environmental applications: performance, generalization, and task suitability.

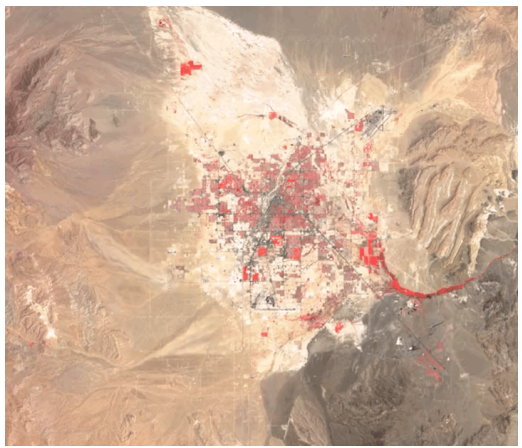
Critically assess research on geospatial representations, foundation models, and Earth embeddings.

Conduct a group research project and communicate methods, results, and limitations.

Light gray fill: responses of the 16 participants on 16 October 2026 (10:00–12:00), last vote per person.

Why this course?

The Earth faces global challenges



Urbanization

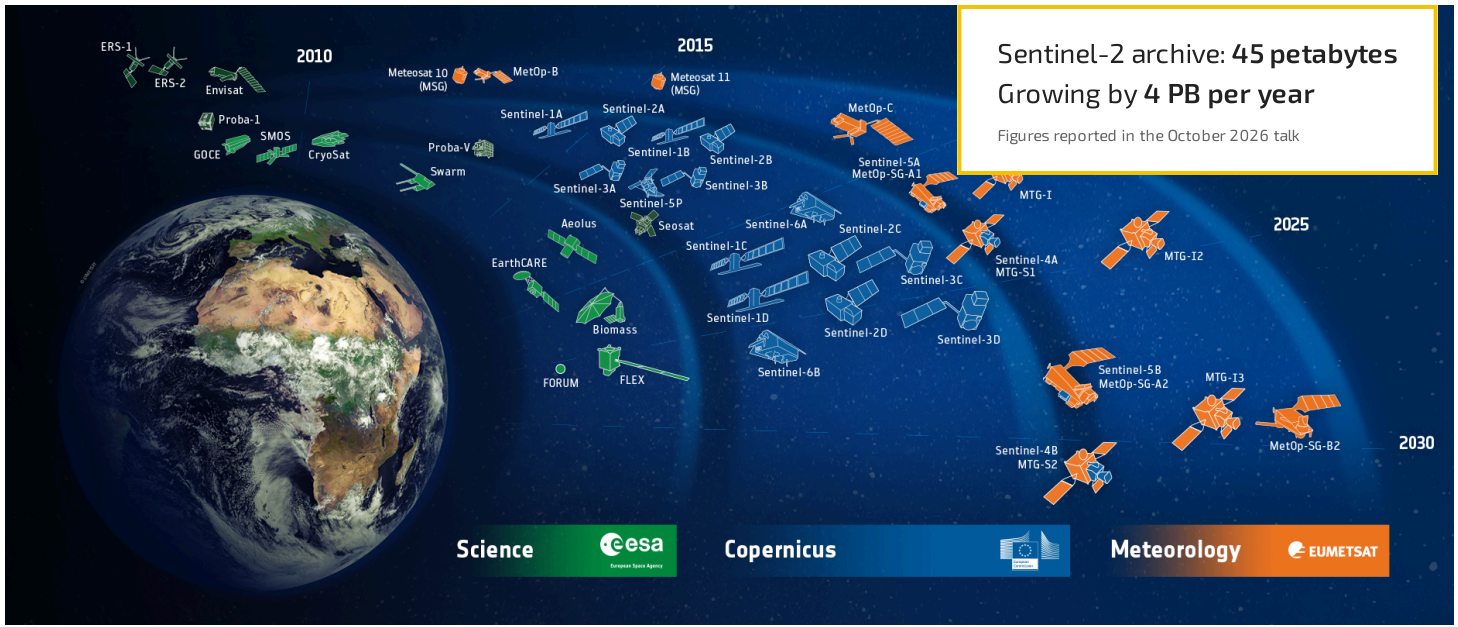
Deforestation

We need to analyze geodata at large scale

to inform policies on biodiversity, food security, human health, and livelihoods.

Images: Google Earth Engine Time Series Gallery · Adapted from the Open Earth Monitor talk, October 2026.

Observing the Earth at unprecedented scales

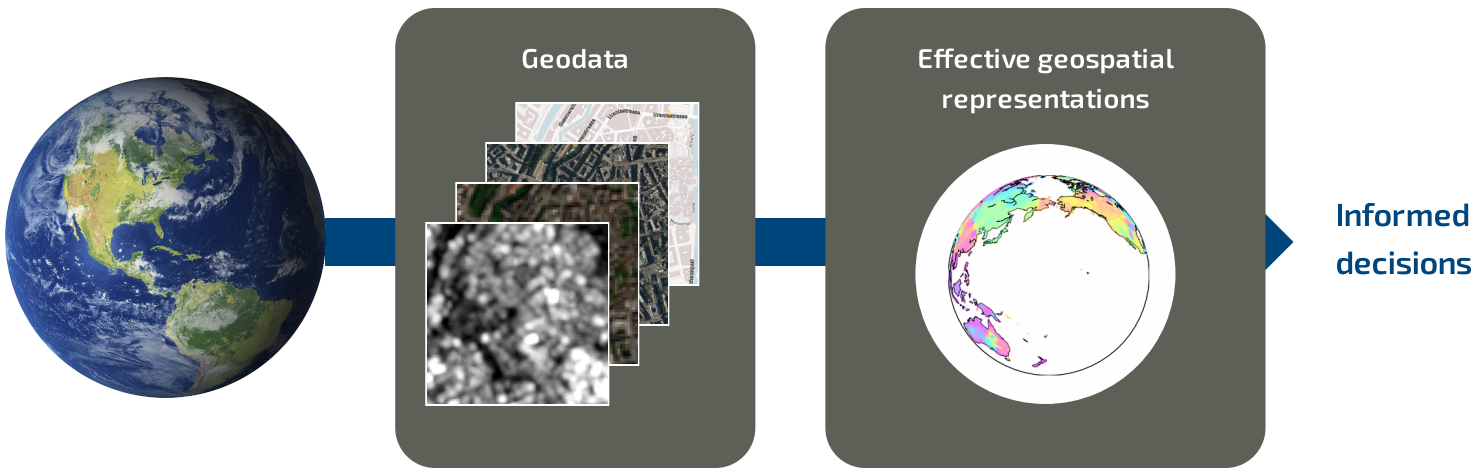


Sentinel-2 archive: **45 petabytes**
Growing by **4 PB per year**

Figures reported in the October 2026 talk

Image: European Space Agency · Archive figures reproduced from the source talk.

We need effective representations of heterogeneous geodata

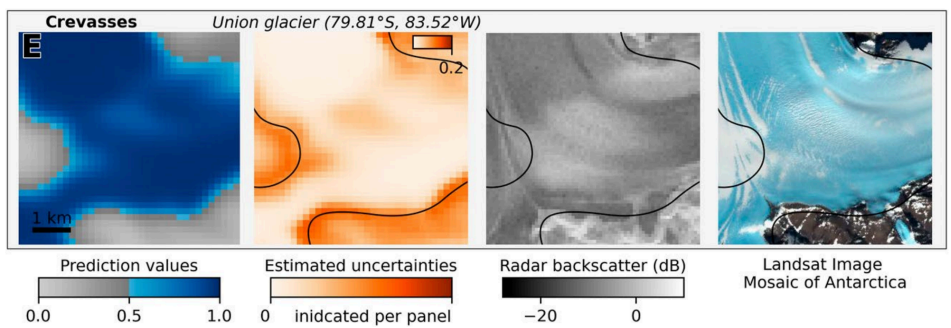


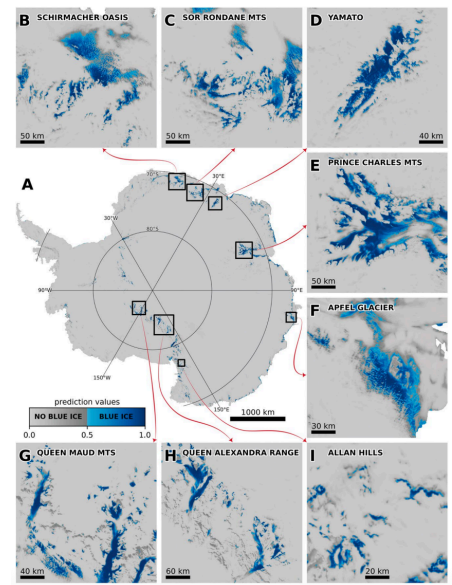
Geospatial data from ground, air, and space at enormous scale

Modern data representations learning with deep neural networks

Finding meteorites in Antarctic blue ice

Veronica Tollenaar
Vrije Universiteit Brussel





Tollenaar, V., Zekollari, H., Pattyn, F., Rußwurm, M., Kellenberger, B., Lhermitte, S., Izeboud, M., & Tuia, D. (2024). Where the White Continent Is Blue: Deep Learning Locates Bare Ice in Antarctica. *Geophysical Research Letters*, 51(3), e2023GL106285. doi:10.1029/2023GL106285

Pollution monitoring

Cuttings Beach, Durban, South Africa
Image: Lisa Guastella

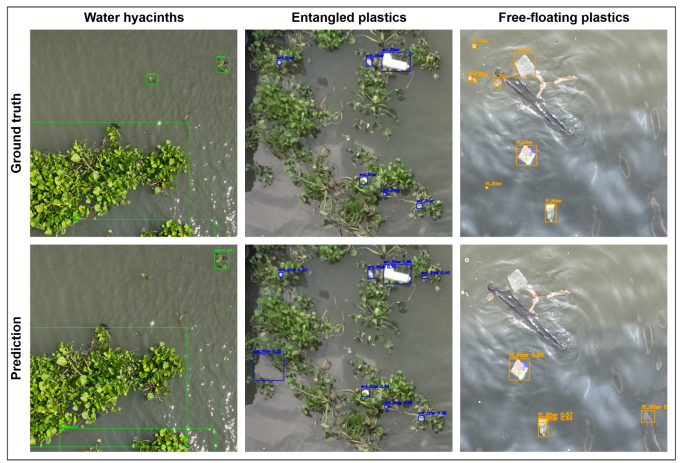


Figure 3: Examples of predicted bounding boxes compared to hand-annotated (ground truth) bounding boxes.

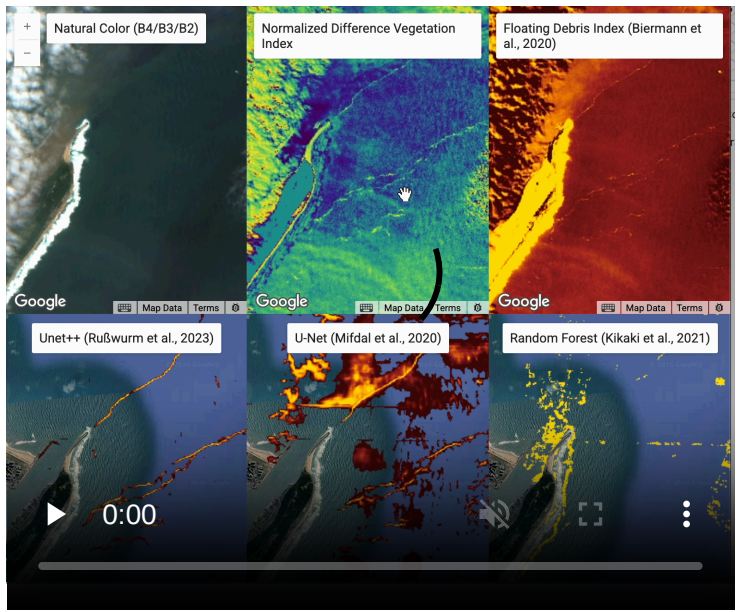
Giel Hagenbeek · Wageningen University & Research

Hagenbeek, G. W., van Emmerik, et al., including Rußwurm, M. (2026). Exploring transferability of plastic-water hyacinth interaction and detection in rivers. *iScience*, 29(6).

Detecting litter in open waters

Cuttings Beach, Durban · Image: Lisa Guastella

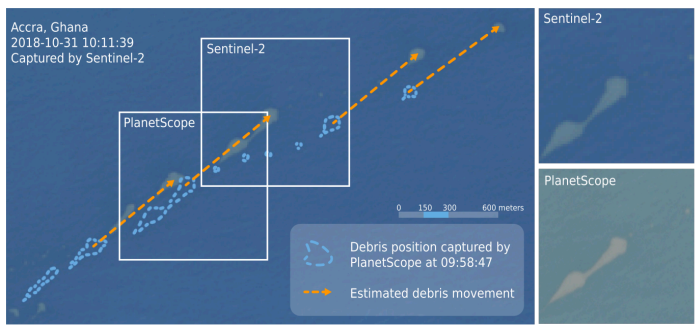
Sentinel-2 satellite

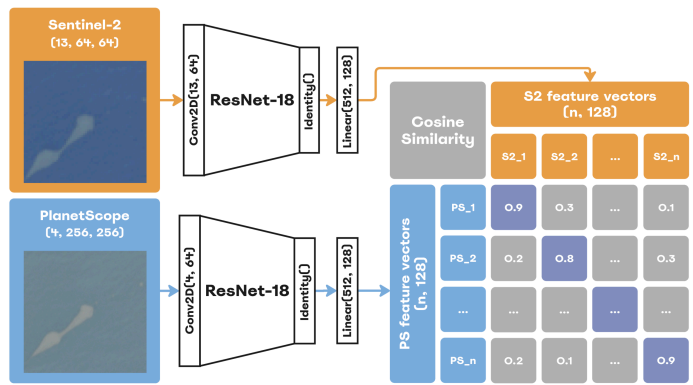


Rußwurm, M., Venkatesa, S. J., & Tuia, D. (2023). Large-scale detection of marine debris in coastal areas with Sentinel-2. *iScience*, 26(12). Marine debris detector

Tracking litter patches across sensors

Gabrielė Tijūnaitytė · Wageningen, Bonn



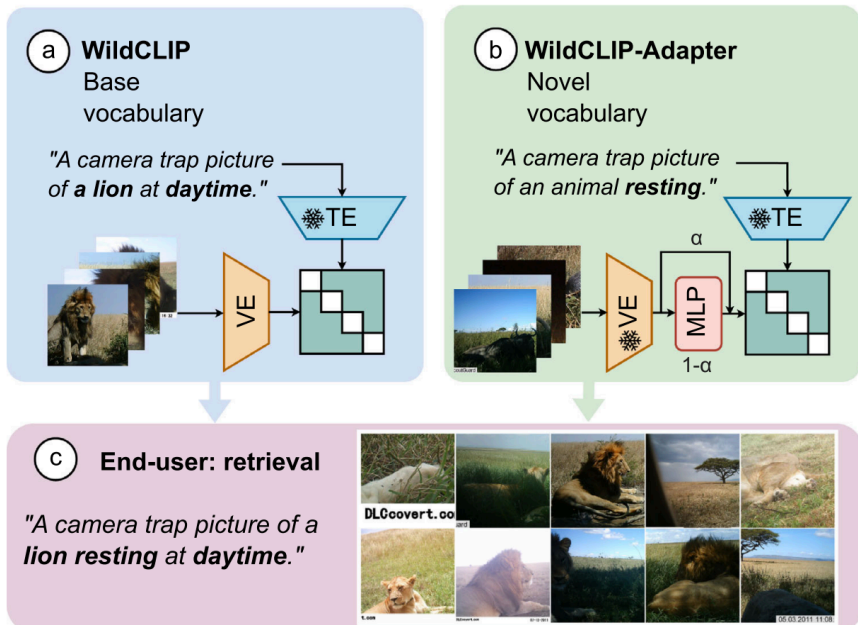


Tijūnaitytė, G., Dalsasso, E., & Rußwurm, M. (2026). Cross-Sensor Matching of Marine Litter Patches in Satellite Double Acquisitions. ECCV Marine Vision Workshop (MaVi).

Text-based image retrieval on camera traps

Trained on the Snapshot Serengeti camera-trap dataset (Swanson et al., 2015) with more than **seven million images**.

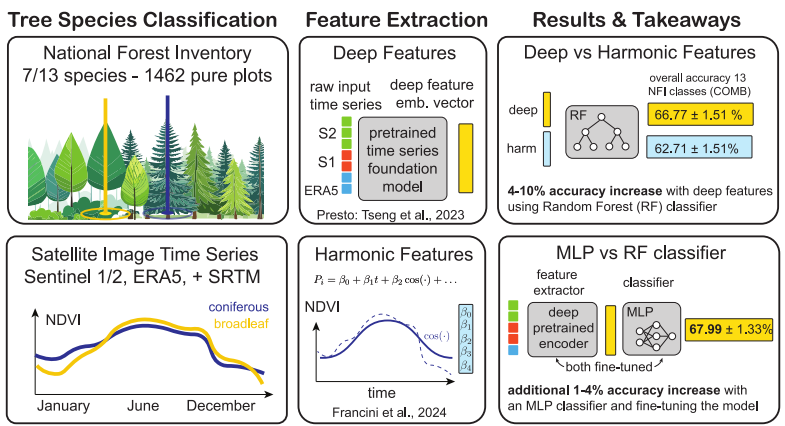
Valentin Gabeff · EPFL

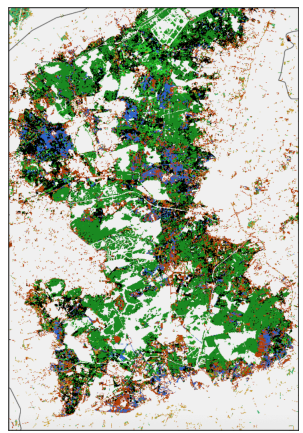


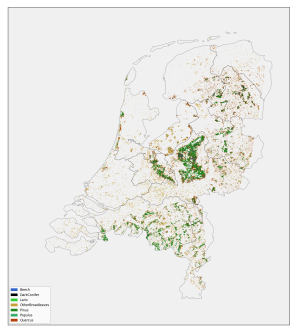
Gabeff, V., Rußwurm, M., Tuia, D., & Mathis, A. (2024). WildCLIP: Scene and animal attribute retrieval from camera trap data with domain-adapted vision-language models. *International Journal of Computer Vision*, 132(9), 3770–3786.

Tree species classification

Takayuki Ishikawa · WUR MSc student





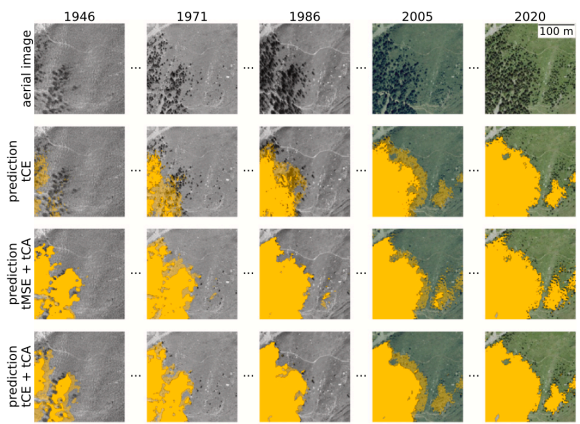


Ishikawa, T., Bonannella, C., Lerink, B. J. W., & Rußwurm, M. (2026). Assessing the effectiveness of deep embeddings for tree species classification in the Dutch Forest Inventory. *Science of Remote Sensing*, 14, 100498. doi:10.1016/j.srs.2026.100498

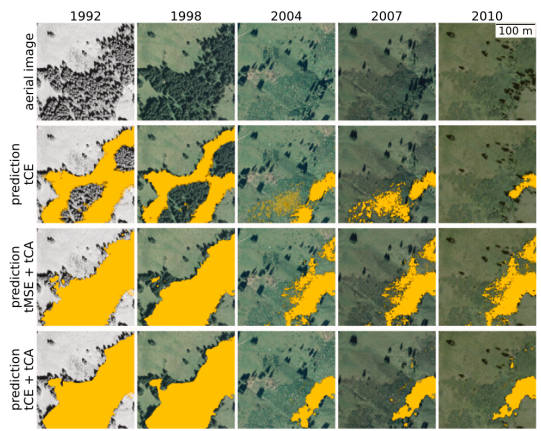
Forest change in the Swiss Alps

Historic aerial photos (1946–2020) reveal forest gains at high altitudes as the tree line advances upwards with a warming climate.

Historic aerial imagery
2,660 tiles
1 km × 1 km
12–20 time steps



(a) Forest gain



(b) Forest loss

Thien-Anh Nguyen · EPFL · Research example reproduced from the source talk, slide 9.

See you next Friday for Lecture 1!

Representations: What is a representation?

We will explore how images, words, and places can be represented.

10:00–12:00 (s.t.) · HS VIII
Meckenheimer Allee 172

Oh god, where did I sign up? — I am excited about the course!

Click on the line to vote.

Light gray fill: responses of the 13 participants on 16 October 2026 (10:00–12:00), last vote per person.